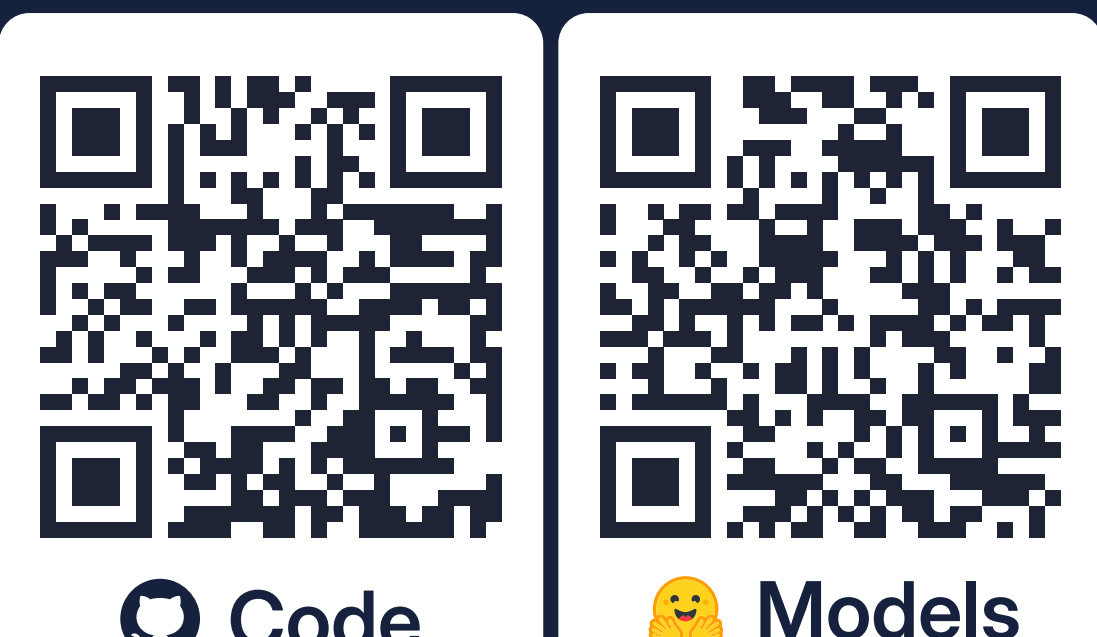


Temporally-Resolved Token Attribution Reveals the Generation Dynamics of Diffusion Language Models

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NeurIPS 2026 • NNArtifacts Workshop

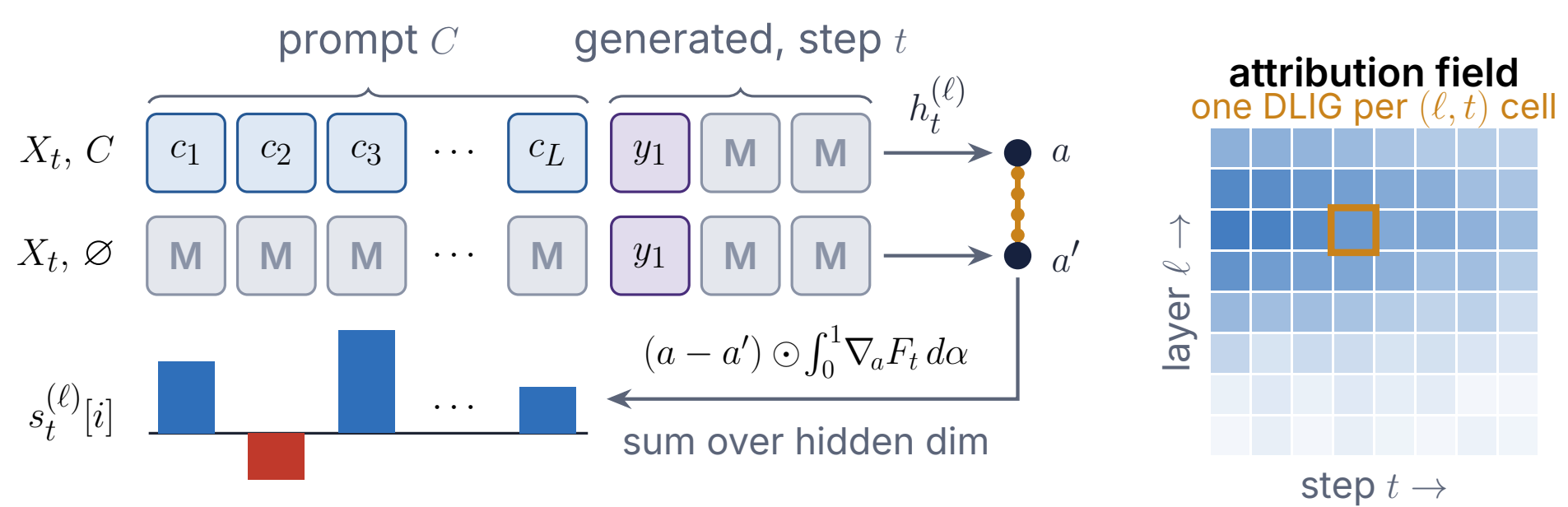


1 Explain where & when

- Diffusion LMs unmask over T steps: support for an output is spread over network **depth** and denoising **time**.
- Token attribution has no notion of a step; DLM interpretability relies on **costly interventions** such as causal tracing [3].

DLIG: Integrated Gradients [1] at any (layer ℓ , step t), giving a per-token attribution field over depth $\times$ time.

2 Diffusion Layer Integrated Gradients



$$\text{SCORE } F_t = \sum_{i=1}^n m_i \log \text{softmax}(z_t^{(L+i)})_{y_i}$$

$$\text{DLIG } \text{DLIG}_t^{(\ell)} = (a - a') \odot \int_0^1 \frac{\partial F_t}{\partial a} (a' + \alpha(a - a')) d\alpha$$

$$\text{PER TOKEN } s_t^{(\ell)}[i] = \sum_j \text{DLIG}_t^{(\ell)}[i, j]$$

- y : the model's own committed tokens (*self-generated*) or a *fixed target*; a' masks the prompt.
- $s > 0$ **drives** the target, $s < 0$ **suppresses** it.
- Exact equivalence**: DLIG = IG on $G_t^{(\ell)}$, which overrides layer ℓ and recomputes the layers above.

3 IG axioms carry over

- Completeness**: $\sum \text{DLIG} = F_t(X_t, C) - F_t(X_t, \emptyset)$ exactly.
- Implementation invariance**: depends only on $G_t^{(\ell)}$.
- Linearity**: $\text{DLIG}(y^+) - \text{DLIG}(y^-)$ attributes the contrast score.
- Symmetry preservation**: straight-line path.

✓ VERIFIED IN PRACTICE

Completeness holds at all 72 tested (layer, step) cells. DiffuGPT-M [2] (355M), $T=64$ steps, $m=12$.

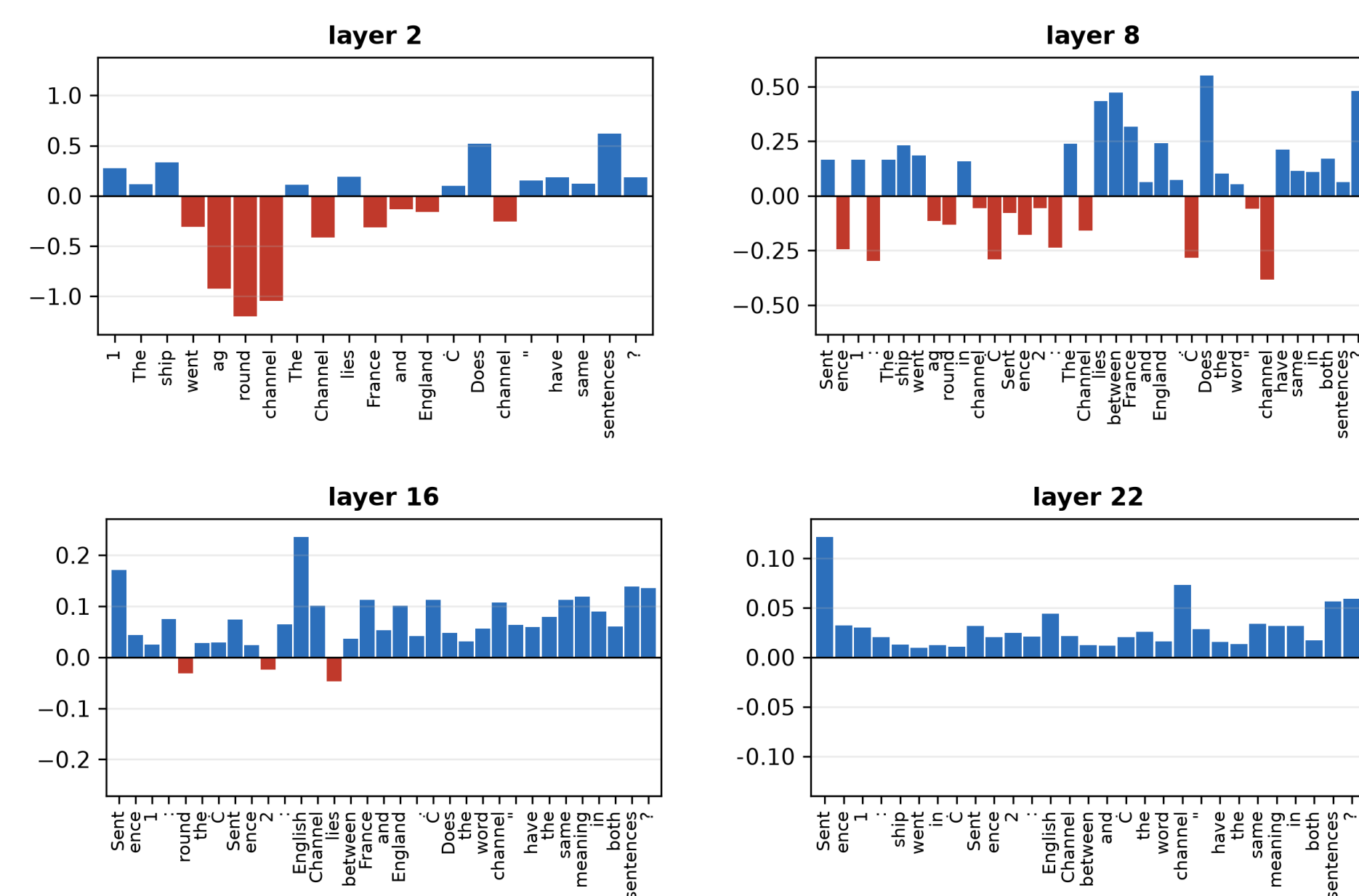
TASK 1 · WORD-SENSE DISAMBIGUATION · WiC

4 Why is the model Yes-biased?

Does a word keep its sense across two sentences [4]? The finetuned model over-answers Yes: **90.8%** accuracy on same-sense vs. **36.4%** on different-sense pairs.

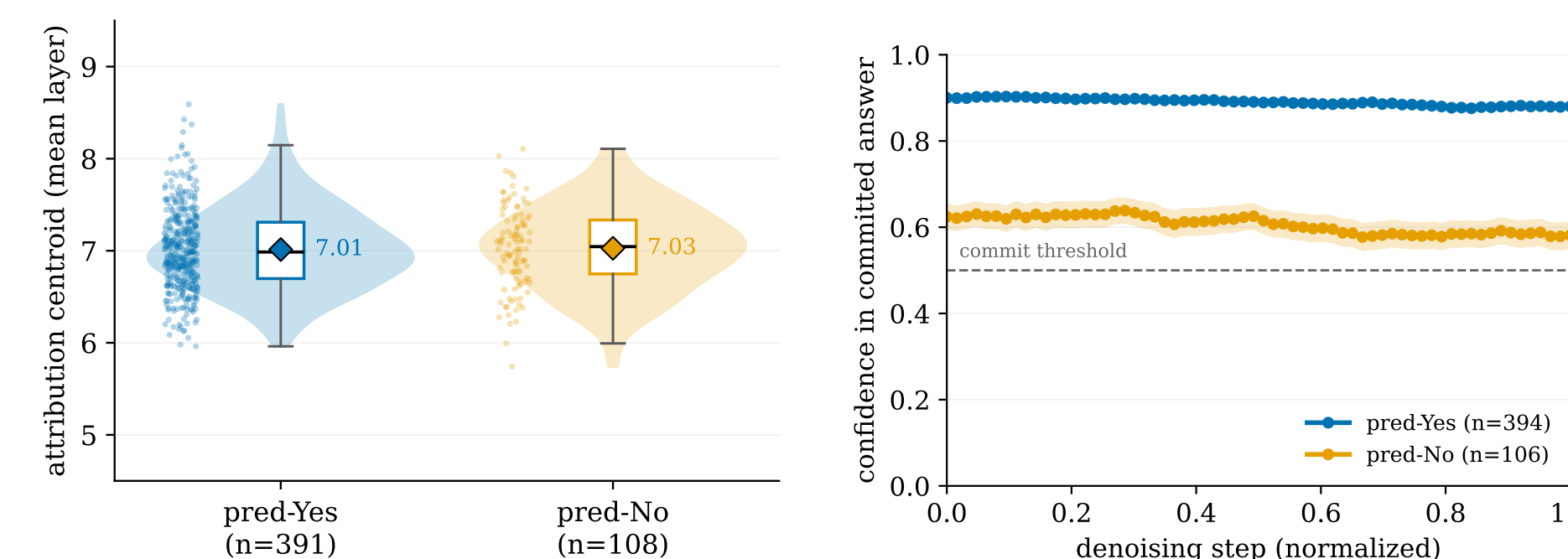
HYPOTHESIS TESTED

Yes and No read prompt evidence at different depths.



One example (pivot word *channel*), selected layers. DLIG averaged over 64 steps; **blue** drives the committed Yes, **red** suppresses it. Large, mixed-sign support in shallow layers thins into answer-aligned support with depth.

Two axes: **attribution depth** (mass-weighted mean layer) and **commitment timing** (answer confidence across steps).



(a) Attribution depth

(b) Commitment timing

- Same depth**: both centre just above layer 7 of 24 (AUC **0.503**, $p = 0.92$).
- Different commitment**: Yes is confident from step 1 and holds; No barely clears the 0.5 threshold (AUC **0.342**, $p < 10^{-4}$).

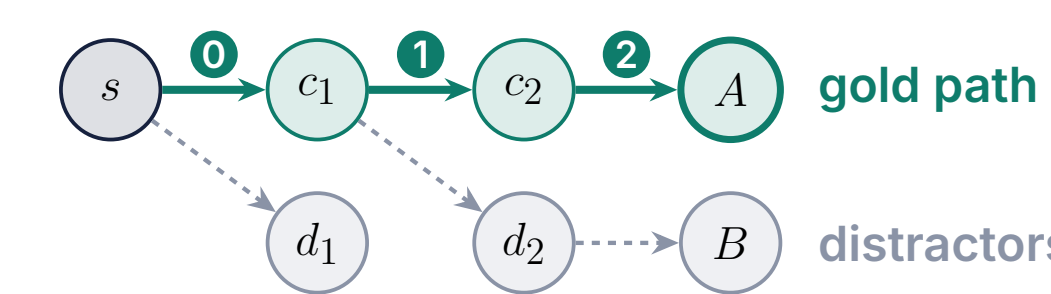
✗ DLIG VERDICT: RULED OUT

The Yes-bias lives in **commitment dynamics**, not in where the evidence is read.

TASK 2 · MULTI-HOP REASONING · ProsQA

5 Do errors look in the wrong place?

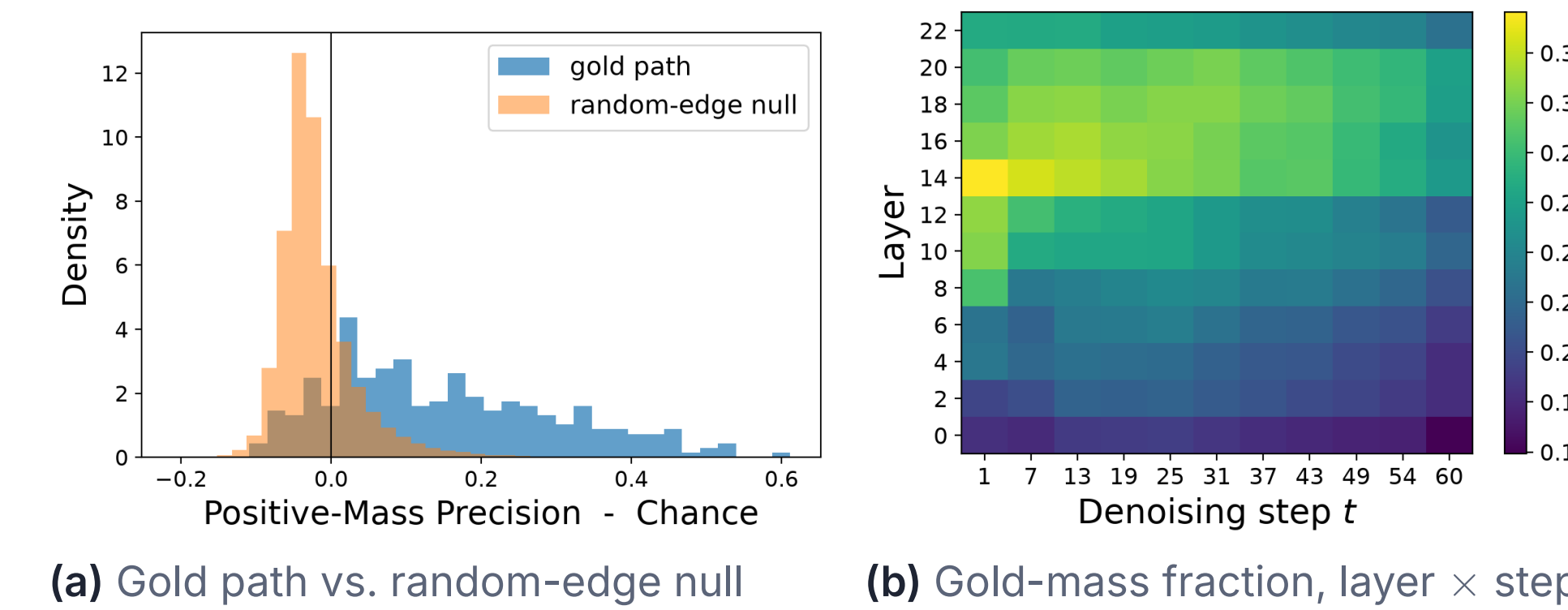
A fact graph with distractors asks "Is s an A or a B ?" [5]. We compare **286** successes with **69** failures.



HYPOTHESIS TESTED

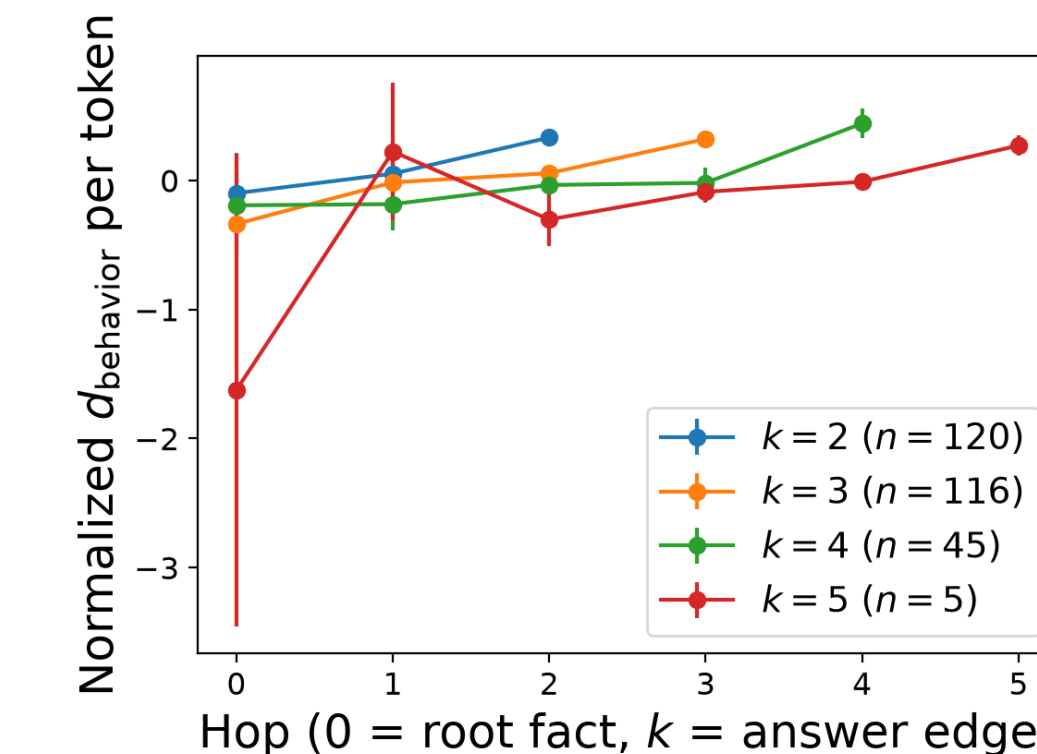
Failures arise from answer support mislocalized onto distractor facts.

Linearity gives a contrastive score $d[i] = s[i](y^+) - s[i](y^-)$ (gold vs. wrong option). **Path precision**: the share of positive d on gold-path tokens.



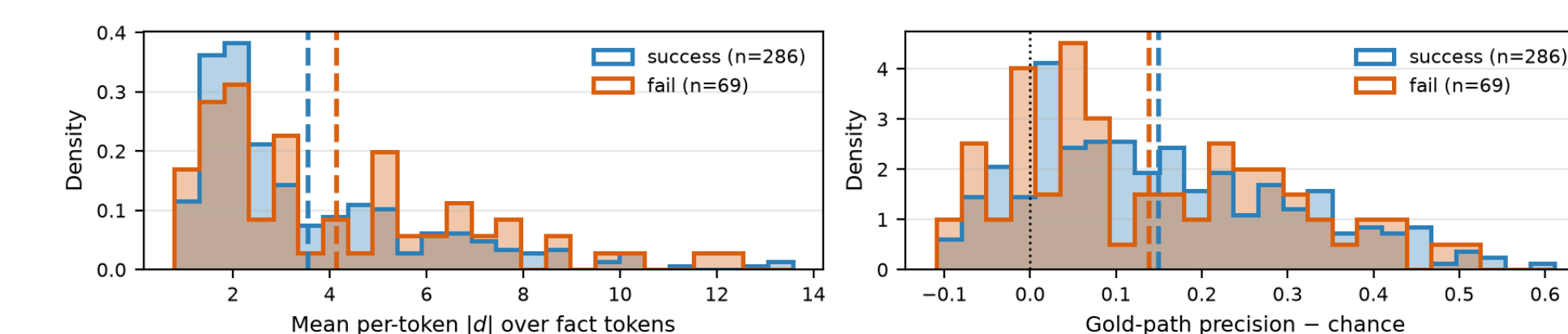
(a) Gold path vs. random-edge null

(b) Gold-mass fraction, layer $\times$ step



(c) Support per gold-path hop

Support sits on the reasoning path: precision is **+0.150** over chance (null 95th pct. $+0.075$; $p < 10^{-15}$), peaks in mid-to-upper layers at early steps, and rises hop by hop toward the answer edge.



(d) Magnitude (left) vs. location (right) by outcome.

Errors keep the localization: precision is **+0.150** for successes vs. **+0.138** for failures (AUC 0.524, n.s.); on failures, mass tracks the model's own cited chain.

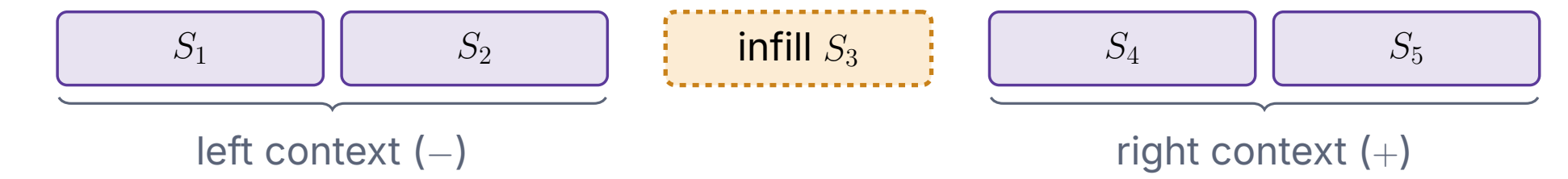
✗ DLIG VERDICT: RULED OUT

The model finds the right fact chain even when it answers wrongly: errors come from **how** it uses the facts, not **where** it looks.

TASK 3 · SENTENCE INFILLING · ROCStories

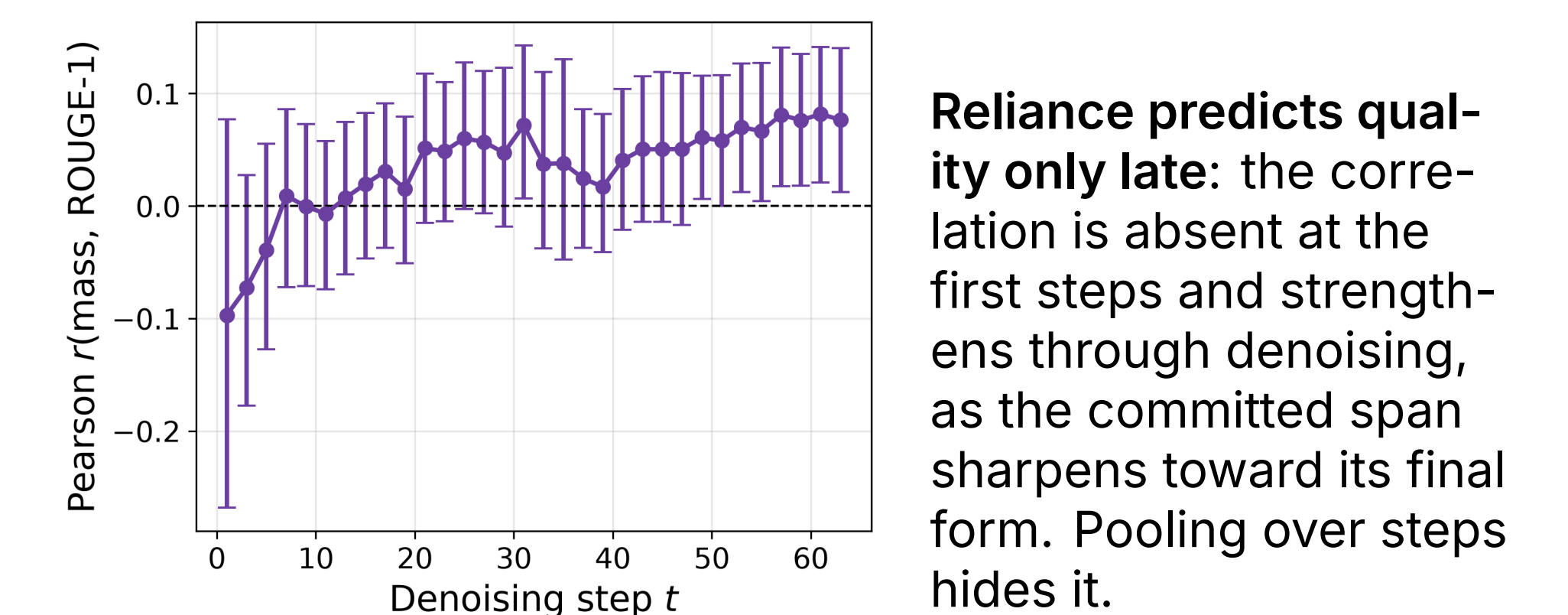
6 When does context start to matter?

Infill sentence 3 of a five-sentence story [6] from both sides; quality is ROUGE-1 against the gold sentence (1,000 stories).

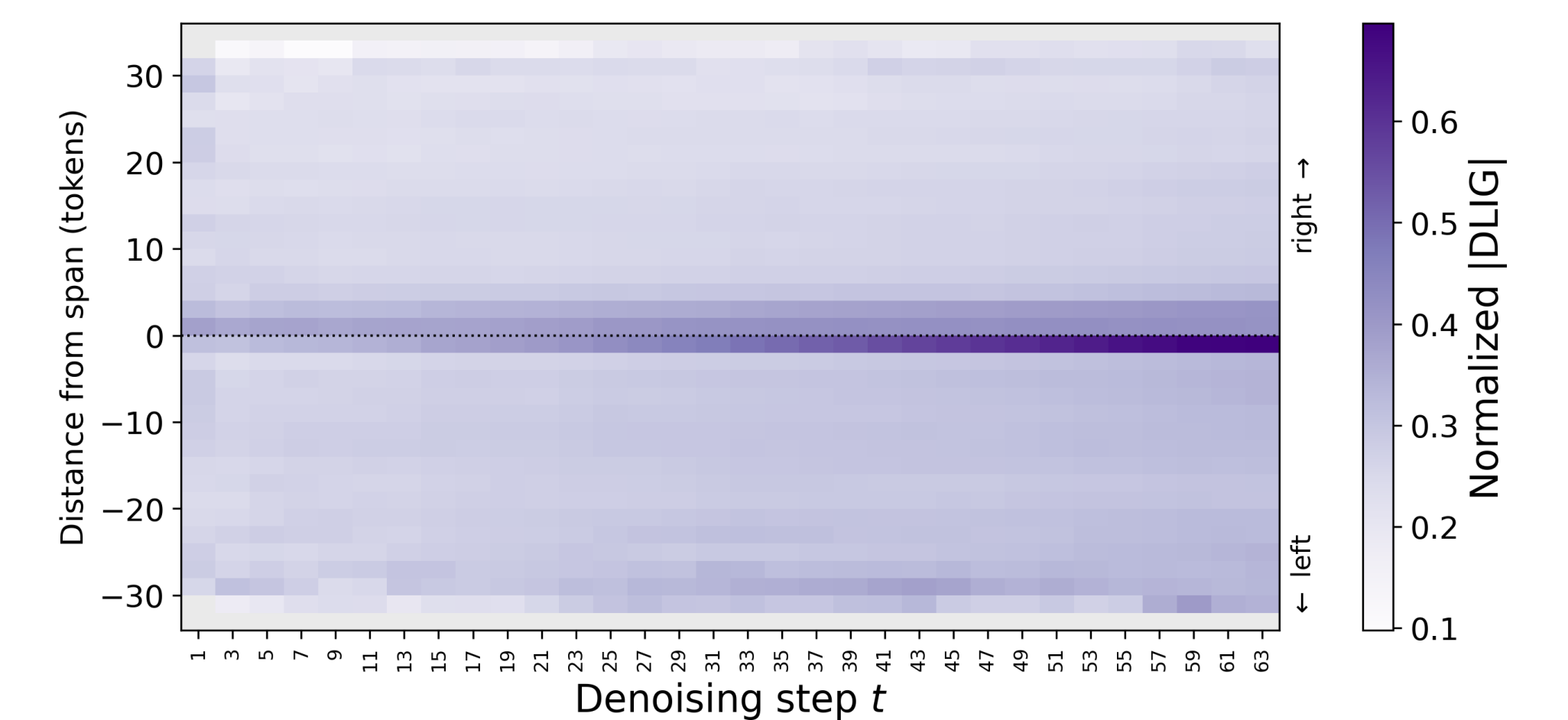


HYPOTHESIS TESTED

Successful infilling relies increasingly on context as the span resolves.



(a) Reliance-quality correlation per step



(b) Attribution vs. signed distance to the span, per denoising step.

- Bidirectional, left-skewed**: mass peaks at both span boundaries; left $>$ right in **57.8%** of story-step cells ($p = 1.9 \times 10^{-204}$). The model uses right context an autoregressive LM cannot see.
- Strengthens late**: context mass grows as the span commits; its amount and direction alone are quality-neutral.

✓ DLIG VERDICT: SUPPORTED

Context becomes predictive of infill quality **only late** in denoising, visible only with temporal resolution.

CONCLUSION

A DLM arrives at its output gradually, so explaining it means reading the **whole denoising trajectory**, not just its end.

WHAT DLIG OFFERS

- Per-token, axiom-preserving attribution at any (layer, step).
- A cheap first check: here it **ruled out two hypotheses** and **supported one**.
- A complement to activation patching and causal tracing.

LIMITATIONS & NEXT STEPS

- Gradient sensitivity is not a causal circuit: read outputs as hypotheses to validate.
- One 355M masked DLM; next: continuous, hybrid and larger DLMs.